ICT TECHNICIAN L

ICT/OS/IT/CC/01/6

Apply Basic Electronics Skills

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962

WRITTEN ASSESSMENT

Cat 3

TIME: 2 HOURS

90%

INSTRUCTIONS TO CANDIDATE

i. *This paper consists of **TWO** sections: **A** and **B**.*
ii. *Answer **ALL** questions in section A and **ANY THREE** (3) questions in section B.*
iii. *Marks for each question are indicated in the brackets.*
iv. *Candidates are provided with a separate answer booklet*
v. *Do not write on the question paper.*

SECTION A (40 MARKS)

Answer ALL questions in this section

1. A technician is troubleshooting a simple series circuit supplying a light bulb and a resi
Outline a tool that can be used to determine if the circuit is complete. (2 M

2. During a lab session, a student measures the current through a resistor in a circuit and fin
lower than expected. Identify FOUR possible causes related to circuit configuration. (4

3. A diode is connected in a circuit and does not allow current to flow in one direction
semiconductor theory, identify the phenomenon causing this behavior. (

4. A student is asked to identify components in a simple electric circuit. State FOU
electronic components commonly used in circuits.

5. A computer engineer is designing a buffer for data storage. Classify TWO types
suitable for temporary and permanent storage.

6. Convert the decimal number 45 into an 8-bit binary. (4 Marks)

7. Illustrate how a NAND gate can be used to implement an AND function using only logic gates. (4 Marks)

8. An LED circuit uses a 9V supply with a 470Ω resistor. Calculate the expected current through the LED. (3 Marks)

9. (a) A student is asked to trace a printed circuit board (PCB) to locate an open circuit. Highlight a two-step systematic approach they would use. (4 Marks)

(b) Outline why a capacitor in a DC circuit initially acts like a short circuit and later blocks current. (2 Marks)

10. Calculate the binary equivalent of hexadecimal 3F. (3 Marks)

11. During a practical test, a student measures voltage drops across resistors in a series circuit. Calculate the expected voltage across a 100Ω resistor in a 12V circuit with 3 resistors: 100Ω, 200Ω, and 300Ω. (4 Marks)

SECTION B (60 MARKS)

Answer THREE questions in this section

12. a) A trainee is required to work with different number systems while developing software for a digital calculator. Explain FIVE disadvantages of using Binary Coded Decimal (BCD) in computers for arithmetic operations. (10 Marks)

b) The rapid development of electronics components has changed the face of the electronic world over the past few years. Explain FIVE emerging trends of such modern electronic components. (10 Marks)

13.

a) Explain THREE ways number systems are applied in computer systems. (6 Marks)

b) Perform the following number systems conversion

156_{10} to:

i. Binary (3 Marks)

ii. Octal (2 Marks)

c) Perform the following number systems arithmetics

i. $11011_2 + 10110_2$ (2 Marks)

ii. $1101_2 - 1011_2$ using 2's Complement (4 Marks)

iii. $101_2 \times 11_2$ (3 Marks)

14. An ICT technician is servicing several computers with suspected power-related faults.

a) Explain how voltage, current, and resistance measurements are used together to diagnose electronic problems. (6 Marks)

b) Describe FOUR risks of incorrect use of electronic measuring instruments in ICT maintenance. (8 Marks)

c) Propose THREE corrective actions to ensure accurate electronic measurements in future tasks. (6 Marks)

15.

a) Outline THREE differences between N-type and P-type semiconductors. (6 Marks)

b) Explain the following terms as used in Bipolar transistor configuration. (6 Marks)

i. Common Base Configuration

ii. Common Emitter Configuration

iii. Common Collector Configuration

c) Discuss FOUR types of Read Only Memory. (8

1. Multimeter

2. High resistance
   Low resistance voltage
   Loose connections
   Incorrect circuit configuration

3. The phenomenon is the formation and widening of the depletion region, which creates a potential barrier that restricts current flow.

4. Resistor
   Capacitor
   Diode
   Transistor

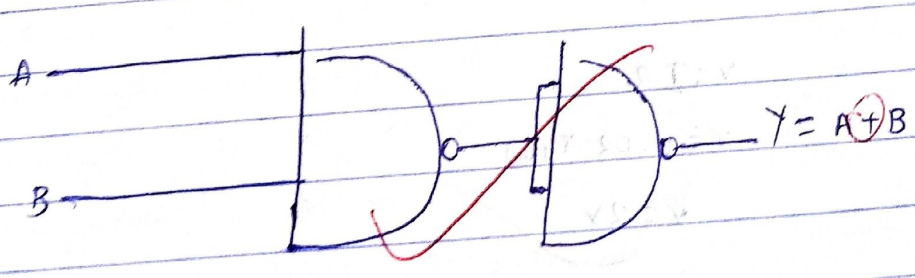
5. Random Access Memory (RAM)
   Read Only Memory.

6. $45 \div 2 = 22$ rem 1
   $22 \div 2 = 11$ rem 0
   $11 \div 2 = 5$ rem 1
   $5 \div 2 = 2$ rem 1
   $2 \div 2 = 1$ rem 0
   $1 \div 2 = 0$ rem 1

   $45_{10} = 101101_2$

   $= 00101101_{BCD}$

7.



A NAND gate produces the opposite output of an AND gate.

8. $I = \frac{V}{R}$

$V_R = 9 - 2 = 7V$

$$I = \frac{7}{470}$$

$$I = 0.01489\ A$$

$$I = 0.01489 \times 1000$$

$$= 14.9\ mA$$

9. Visual Inspection.
Continuity testing.

b) Initially, an uncharged capacitor allows current to flow as it charges, behaving like a short circuit.
As the capacitor charges, the voltage across it increases.
When fully charged, it blocks steady DC current and behaves like an open circuit.

10. 3 = 0011
F = 1111

$$3F_{16} = 0011\ 1111_2$$

$$= 00111111$$

11. $R_T = R_1 + R_2 + R_3$

$R_T = 100 + 200 + 300$

$R_T = 600\Omega$

$$I = \frac{V}{R}$$

$$I = \frac{12}{600} \quad I = 0.02\ A$$

$$V = IR$$

$$V = 0.02 \times 100$$

$$V = 2V$$

SECTION B

12) It wastes memory space - BCD uses 4 bits to represent each decimal digit, making it less efficient than pure binary.
Slow arithmetic operations - arithmetic operations in BCD are more complicated and may take more time.
Complex hardware - special circuits are required to perform BCD arithmetic operations.
Limited number representation - BCD has unused bit combinations, and since its only 10 out of 16 possible combination represent valid decimal digits.
It requires correction - after BCD addition, correction may be required.

13) Internet of Things (IoT) - components are being designed to connect devices to the internet for communication and data sharing.
Artificial Intelligence - Modern electronic components are incorporating AI capabilities to perform intelligent tasks.
Energy efficiency - components are being developed to consume less electrical power.
Flexible electronics - the electronic components are being designed to bend and flex making them useful in wearable devices.
Miniaturization - electronic components are becoming smaller, allowing the development of compact devices.

14a) Voltage measurement - a voltmeter is used to check the voltage supplied to components.
Current measurement - an ammeter is used to measure current flowing through a circuit.
Resistance measurement - an ohmeter is used to measure resistance and identify open circuits, short circuits or faulty components.

b) Electric shock - Incorrect handling of measuring instruments can expose technicians to dangerous electrical voltage.
Damage equipment - Incorrect connections can damage the measuring instrume and electronic components.
Inaccurate readings - Incorrect measurements can lead to wrong diagnosis of ele faults.

Short circuits - Incorrect connection of probes can cause short circuits.

14c) Proper training - Technicians might be trained the correct use of measuring instruments.

Regular calibration - calibrate measuring instruments regularly to maintain accuracy.

Correct instrument settings - Select the correct measurement mode, range before taking any measurements.

15a) N-type semiconductor is formed by adding pentavalent impurities while P-type is formed by adding trivalent impurities.

N-type semiconductors the electrons are the majority charge carriers while P-type semiconductor ~~are~~ electrons the holes are the majority charge carriers.

b) Common Base Configuration - a transistor configuration where the base terminal is common to both inputs and output circuits.

Common Emitter Configuration - a transistor configuration where the emitter terminal is common to both the input and output circuits.

Common Collector Configuration - a transistor configuration where the collector terminal is common to both the input and output circuits.

c) Mask Read Only Memory - Memory programmed by the manufacturer during production.

Programmable Read Only Memory - this is a memory that can be programmed once by the user.

Erasable Programmable Read Only Memory - the memory whose contents can be erased using ultraviolet light.

~~EEP~~ Electrically Erasable Programmable Read Only Memory - memory whose contents can be erased and reprogrammed electrically.